Regression analysis

Ml Linear regression

Regression analysis is a statistical method used to examine the relationship between one dependent variable and one or more independent variables. The goal of regression analysis is to understand how the dependent variable changes when one or more independent variables are altered, and to create a model that can predict the value of the dependent variable based on the values of the independent variables.

Steps in regression analysis:

1. **Define the research question**: Identify the dependent variable and the independent variable(s).
2. **Collect and prepare data:** Gather data in a tabular format, clean and process it for analysis.
3. **Visualize the data:** Use scatter plots to explore relationships and spot trends or outliers.
4. **Check assumptions:** Make sure the model assumptions such as linearity, independence, and error normality hold for your data.
5. **Fit the linear regression model:** Use statistical software to calculate regression coefficients (intercept and slope) that minimize the sum of squared residuals.
6. **Interpret the model:** Analyze the estimated coefficients, their standard errors, t-values, and p-values to determine the statistical significance of relationships. Use R-squared values to evaluate goodness-of-fit.
7. **Validate the model:** Split data into training and testing sets to measure predictive accuracy using metrics such as mean squared error.
8. **Report results:** Summarize findings, interpretations, limitations and assumptions in a clear manner.

Why are machine learning problems considered statistical inference problems ??

Machine learning is considered a statistical inference problem because ML algorithms learn relationships, estimate parameters, and make predictions about the population using a limited sample of data.

This is the same goal as statistical inference.

Let suppose
lpa=f(cgpa,iq)+ε
Here:

```
- lpa is the dependent variable.
- f(gpa,iq) represents a function of the independent
variables gpagpa and iqiq.
- ε is the error term or residual.
```

so Linear regression equation for **f(cgpa,iq)** will be :
$lpa=\beta0+\beta1gpa+\beta2iq$

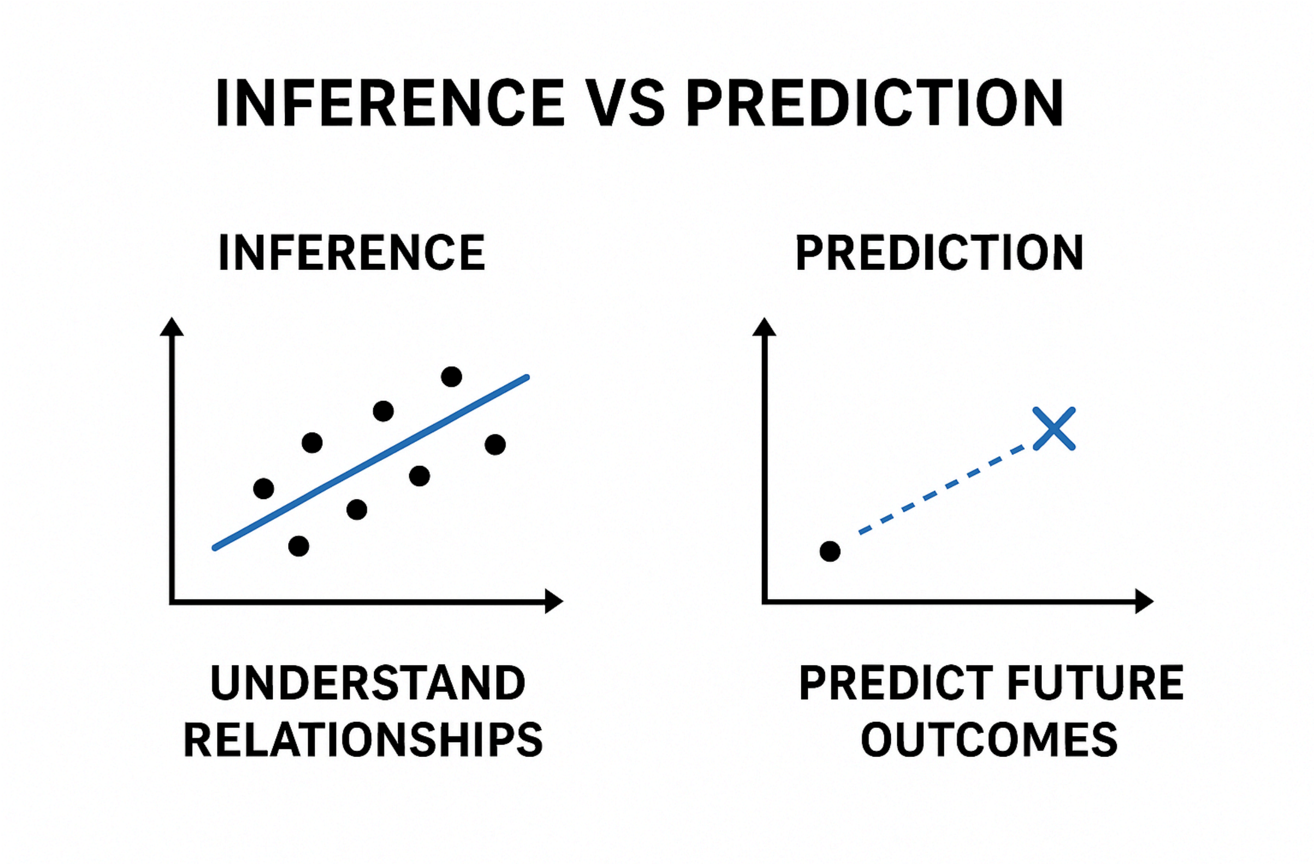
but ,
When estimating the parameters β0,β1,β2of a regression model using sample data, we obtain estimates b0,b1,b2b0,b1,b2 instead of the true population values β0,β1,β2. These estimates may differ from the true values due to random sampling fluctuations.

So equation,
lpa=f′(cgpa,iq)+reducible error+ε
Here:

- lpa is your target variable.
- f′(cgpa,iq) is an estimated function based on sample data.
- reducible error represents the part of error that can be minimized through better modeling or using more data.
- ε is the irreducible error (random noise that cannot be explained by the model).

Why Regression Analysis?

- It provides estimates for model parameters, enabling both explanation and forecasting.
- Enables assessment of how much each variable contributes to the outcome.
- Supports decision-making and actionable insights based on data.



Note :

Regression is necessary because it serves two essential purposes: it provides statistical understanding (inference) by explaining and quantifying relationships between variables, and it enables practical forecasting (prediction) by estimating future outcomes based on these learned relationships.

Statsmodel Linear Regression

X --> Y is there any relationship

- it's Linear ?
- is strong ?
- what is indivable(x1 , x2 , x3) relation with y

```
import pandas as pd
import statsmodels.api as sm

# Load the dataset
url = "https://raw.githubusercontent.com/justmarkham/scikit-learn-
videos/master/data/Advertising.csv"
data = pd.read_csv(url, index_col=0)

# Define the independent variables (add a constant for the intercept)
X = data[['TV', 'Radio', 'Newspaper']]
X = sm.add_constant(X)
```

```
# Define the dependent variable
y = data['Sales']

# Fit the model using the independent and dependent variables
model = sm.OLS(y, X).fit()

# Print the summary of the model
print(model.summary())
```

Output :

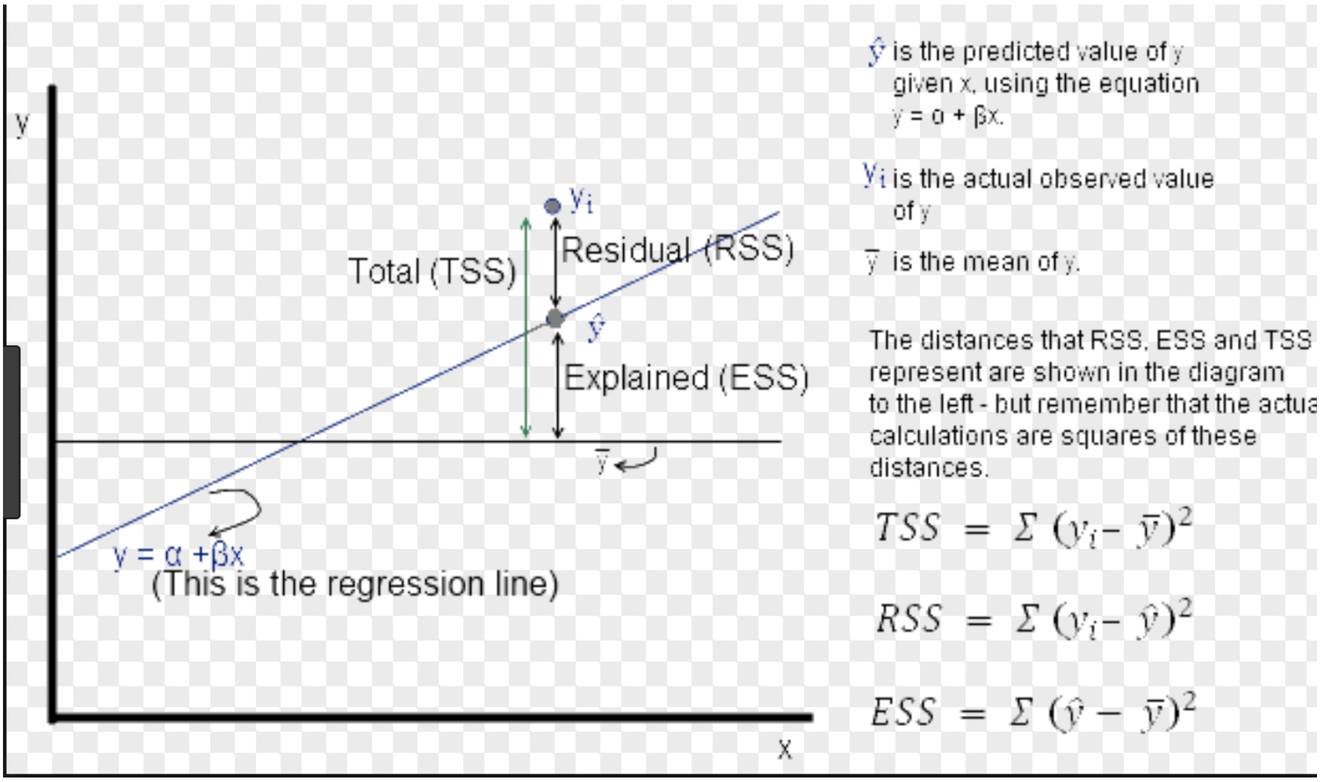
```
                            OLS Regression Results
==============================================================================
Dep. Variable:                  Sales   R-squared:                       0.897
Model:                            OLS   Adj. R-squared:                  0.896
Method:                 Least Squares   F-statistic:                     570.3
Date:                Sat, 29 Apr 2023   Prob (F-statistic):           1.58e-96
Time:                        07:32:56   Log-Likelihood:                -386.18
No. Observations:                 200   AIC:                             780.4
Df Residuals:                     196   BIC:                             793.6
Df Model:                           3
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const          2.9389      0.312      9.422      0.000       2.324       3.554
TV             0.0458      0.001     32.809      0.000       0.043       0.049
Radio          0.1885      0.009     21.893      0.000       0.172       0.206
Newspaper     -0.0010      0.006     -0.177      0.860      -0.013       0.011
==============================================================================
Omnibus:                       60.414   Durbin-Watson:                   2.084
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              151.241
Skew:                          -1.327   Prob(JB):                     1.44e-33
Kurtosis:                       6.332   Cond. No.                         454.
==============================================================================
```

We will start we sub set of the data :

```
R-squared:                       0.897
Adj. R-squared:                  0.896
F-statistic:                     570.3
Prob (F-statistic):           1.58e-96
Log-Likelihood:                -386.18
AIC:                             780.4
BIC:                             793.6
```

Sub Points to understand in depth the above fig

TSS → Total sum of squares
RSS → Residual sum of squares
ESS → Explained sum of squares



Degree of Freedom (DoF)

Degree of freedom = **the number of independent values that are free to vary.**

1. Degrees of freedom for the model (df_model)

Meaning:
This is simply the **number of independent variables (predictors)** in your regression model.
$y = \beta0 + \beta1\ x1 + \beta2\ x2 + \beta3\ x3$
Number of independent variables = 3
df_model = 3

2. Degrees of freedom for the residuals (df_residuals)

This tells you how many **independent pieces of information remain** to estimate the variability of the residuals (errors) *after fitting the model*.

df_residuals = n − (k+1)

Where:

- **n = number of rows (observations)**
- **k = number of independent variables**
- **+1 = intercept (b_0)**

F-Test

The F-test is a statistical test that checks whether groups or models are significantly different by comparing their variances.

It is used to answer questions like:

- Do the X variables explain Y? (Regression)
- Are two or more groups different? (ANOVA)
- Are two variances equal? (Variance comparison)

In all cases, the F-test uses a value called the F-statistic.

Breakdown in Simple Steps

F-Test (whole process):
1. Set null hypothesis
2. Set alternative hypothesis
3. Fit regression model
4. Calculate F-statistic (this is ONE step)

F = MSE/MSR

5. Compare with distribution
6. Find p-value
7. Make conclusion (significant or not)

Part	Meaning
F-Statistic	Your score in the exam (e.g., 12.4)
F-Test	The rule that decides whether you pass or fail
Prob(F-Statistic)	The chance that your score is high enough to pass (p-value) chance your score is real

T-Test

A **t-test** is a statistical test used to check whether the **difference between two values (or groups)** is **real or just due to random chance**.

Use of t-test in Machine Learning

1. To test significance of individual features (coefficients) in linear regression

While the **F-test** checks the **overall model**, the **t-test** checks **each variable separately**.

2. In Linear Regression

To check if an **individual feature (X_1, X_2, …)** actually affects the target (Y).

Performing a t-test for a simple linear regression, including the intercept term and using the p-value approach, **involves the following steps**:

1. **State the null and alternative hypotheses for the slope and intercept coefficients:**

For the slope coefficient β1:

- Null hypothesis (H0): β1=0
- Alternative hypothesis (H1): β1≠0

For the intercept coefficient β0β0:

- Null hypothesis (H0): β0=0
- Alternative hypothesis (H1): β0≠0.

2. **Estimate the slope and intercept coefficients (b0 and b1):**
   Using the sample data, calculate the slope (b1b1) and intercept (b0b0) coefficients for the regression model.
3. **Calculate the standard errors for the slope and intercept coefficients (SE(b0b0) and SE(b1b1)):**

$$SE(b_1) = \sqrt{\frac{\sum(y_i - \hat{y}_i)^2}{(n-2)\sum(x_i - \bar{x})^2}}$$

$$SE(b_0) = \sqrt{\frac{\sum(y_i - \hat{y}_i)^2}{n-2}}$$

4. **Compute the t-statistics for the slope and intercept coefficients:**

$$t_{b_1} = \frac{b_1 - \beta_1}{SE(b_1)}$$

$$t_{b_0} = \frac{b_0 - \beta_0}{SE(b_0)}$$

Test	Use Case
t-test	Compare **exactly 2 groups** OR test **1 coefficient (β)**
F-test	Compare **2+ groups**, OR check **overall significance of model**

Feature	t-Test	F-Test
What it tests	Tests **one coefficient** at a time	Tests **all coefficients together**
Purpose	Check if a **single feature** affects Y	Check if **model as a whole** explains Y
Null Hypothesis	$\beta_i = 0$ → feature is useless	All β's = 0 → model is useless
Alternative H_1	$\beta_i \neq 0$ → feature important	At least one β ≠ 0 → model significant

Feature	t-Test	F-Test
Used for	**Individual feature significance**	**Overall model significance**
Where used	Linear regression, A/B test	Regression, ANOVA
Output	t-statistic + p-value	F-statistic + p-value
Interpretation	“Is this one variable useful?”	“Is the whole model useful?”
If test fails	Remove/ignore that feature	Model does NOT explain Y
Data comparison	Compares 2 groups	Compares ≥2 groups